dictates the rhythm of the cosmic dance that is tirelessly and meticulously executed by billions upon billions of cosmic inhabitants, from asteroids to planets to stars to galaxies. More than three centuries of Newton's influence causes us to take for granted that a single force —gravity—is responsible for this wealth of terrestrial and extraterrestrial happenings. But before Newton there was no understanding that an apple falling to earth from a tree bore witness to the same physical principle that keeps the planets revolving around the sun. With an audacious step in the service of scientific hegemony, Newton united the physics governing both heaven and earth and declared the force of gravity to be the invisible hand at work in each realm.

Newton's view of gravity might be called the great equalizer. He declared that absolutely everything exerts an attractive gravitational force on absolutely everything else. Regardless of physical composition, everything exerts as well as feels the force of gravity. Based on a close study of Johannes Kepler's analysis of planetary motion, Newton deduced that the strength of the gravitational attraction between two bodies depends on precisely two things: the amount of stuff composing each of the bodies and the distance between them. "Stuff" means matter—this comprises the total number of protons, neutrons, and electrons, which in turn determines the mass of the object. Newton's universal theory of gravity asserts that the strength of attraction between two objects is larger for larger-mass objects and smaller for smaller-mass objects; it also asserts that the strength of attraction is larger for smaller separations between the objects and smaller for larger separations.

Newton went much further than this qualitative description and wrote down equations that quantitatively describe the strength of the gravitational force between two objects. In words, these equations state that the gravitational force between two bodies is proportional to the product of their masses and inversely proportional to the square of the distance between them. This "law of gravity" can be used to predict the motion of planets and comets around the sun, the moon about the earth, and rockets heading off for planetary explorations, as well as more earthbound applications such as baseballs flying through the air and divers spiraling poolward from springboards. The agreement